LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Internal ballistics, grain configuration and form function
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Internal ballistics, grain configuration and form function. D Resells energy equation and secondary energy losses, muzzle velocity and peak pressure analysis
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT slides on grain configuration; Instructor supervision required for Q&A;
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four main branches of ballistics?
(b) Define internal ballistics in the context of gunnery.
(c) List the primary factors affecting projectile motion inside the barrel.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Branches of ballistics overview • Internal ballistics definition • Propellant energy equation • Secondary energy losses • Muzzle velocity analysis	L & Ds	PPT, IB	15 min
(b)	• Grain configuration types • Form function concept • Peak pressure analysis • Burn rate characteristics • Pressure curve interpretation	L & Ds	PPT, PDF	20 min
(c)	• Energy loss mechanisms • Gas dynamics in barrel • Muzzle velocity calculation • Pressure peak determination • Efficiency of propellant	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how grain configuration influences the pressure curve.
(b) Describe the relationship between form function and muzzle velocity.
(c) Identify two sources of secondary energy losses in internal ballistics.

12. Home Assignment and Follow Up	Prepare a brief note on how changing the grain shape affects the peak pressure and muzzle velocity, referencing the next lesson on external ballistics.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	External ballistics –motion in vacuum, motion in air, formula for air resistance, ballistic coefficient and limiting velocity	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min